

W5 – Differentiation Rules for Exponential Functions

Unit 3

MCV4U

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SOLUTIONS

1)a) Rewrite the function $y = b^x$ with base e .

$$y = e^{\ln(b^x)}$$

$$y = e^{x \ln(b)}$$

b) Find the derivative of your function in part a) and simplify.

$$y' = e^{x \ln(b)} [\ln(b)]$$

$$y' = e^{\ln(b^x)} [\ln(b)]$$

$$y' = b^x \ln(b)$$

2) Differentiate with respect to x .

a) $y = e^{-3x}$

$$y' = e^{-3x} (-3)$$

$$y' = -3e^{-3x}$$

b) $f(x) = e^{4x-5}$

$$f'(x) = e^{4x-5} (4)$$

$$f'(x) = 4e^{4x-5}$$

c) $y = e^{2x} - e^{-2x}$

$$y' = e^{2x} (2) - e^{-2x} (-2)$$

$$y' = 2e^{2x} + 2e^{-2x}$$

d) $y = 2^x + 3^x$

$$y' = 2^x \ln(2) + 3^x \ln(3)$$

e) $f(x) = 3e^{2x} - 2^{3x}$

$$f'(x) = 3e^{2x} (2) - 2^{3x} \ln(2) (3)$$

$$= 6e^{2x} - 3 \ln(2) (2^{3x})$$

f) $y = 4xe^x$

$$y' = 4(e^x) + e^x (4x)$$

$$y' = 4e^x + 4xe^x$$

g) $y = 5^x e^{-x}$

h) $f(x) = x e^{2x} + 2e^{-3x}$

$$y' = 5^x (\ln 5)(e^{-x}) + (e^{-x})(-1)(5^x)$$

$$y' = 5^x (\ln 5)(e^{-x}) - e^{-x}(5^x)$$

$$y' = 5^x (e^{-x})(\ln 5 - 1)$$

$$f'(x) = 1(e^{2x}) + (e^{2x})(2)x - 6e^{-3x}$$

$$f'(x) = e^{2x} + 2x e^{2x} - 6e^{-3x}$$

$$f'(x) = e^{2x} (1 + 2x - 6e^{-5x})$$

3) Determine the derivative with respect to x for each function.

a) $y = e^{-x} \sin x$

b) $y = e^{\cos x}$

$$y' = -e^{-x} \sin x + \cos x (e^{-x})$$

$$y' = e^{\cos x} (-\sin x)$$

$$y' = e^{-x} (-\sin x + \cos x)$$

$$y' = -\sin x (e^{\cos x})$$

$$y' = e^{-x} (\cos x - \sin x)$$

c) $f(x) = e^{2x}(x^2 - 3x + 2)$

d) $g(x) = 2x^2 e^{\cos(2x)}$

$$f'(x) = 2e^{2x}(x^2 - 3x + 2) + (2x - 3)(e^{2x})$$

$$f'(x) = e^{2x} [2x^2 - 6x + 4 + 2x - 3]$$

$$f'(x) = e^{2x} (2x^2 - 4x + 1)$$

$$g'(x) = 4x [e^{\cos(2x)}] + e^{\cos(2x)} [-\sin(2x)](2)(2x^2)$$

$$g'(x) = 4x e^{\cos(2x)} [1 - x \sin(2x)]$$

4) Identify the coordinates of any local extrema of the function $y = e^x - e^{2x}$

$$y = e^x - e^{2x}$$

$$y' = e^x - 2e^{2x}$$

$$0 = e^x(1 - 2e^x)$$

$$e^x \neq 0 \quad 1 - 2e^x = 0$$

$$1 = 2e^x$$

$$\frac{1}{2} = e^x$$

$$x = \ln\left(\frac{1}{2}\right)$$

First Derivative Test:

	$\ln\left(\frac{1}{2}\right)$	
Test Value	-10	10
$y' = e^x - 2e^{2x}$	+	-
$y = e^x - e^{2x}$	Increasing	decreasing

Local max at $\left(\ln\left(\frac{1}{2}\right), \frac{1}{4}\right)$

5) Find an equation for the tangent to the curve $y = 2e^{2x} + 2x + 1$ when $x = 0$.

Point:

$$y(0) = 2e^{2(0)} + 2(0) + 1$$

$$y(0) = 2 + 0 + 1$$

$$y(0) = 3$$

$$(0, 3)$$

Slope:

$$y' = 4e^{2x} + 2$$

$$y'(0) = 4e^{2(0)} + 2$$

$$y'(0) = 4 + 2$$

$$y'(0) = 6$$

$$m = 6$$

Eqⁿ

$$y = mx + b$$

$$3 = 6(0) + b$$

$$b = 3$$

$$y = 6x + 3$$

6) Find the equation of the tangent to $y = x \ln x$ that is parallel to $y = 3x + 7$.

$$m = 3$$

$$y' = (1) \ln(x) + \frac{1}{x \ln e}$$

$$y = \ln(x) + 1$$

$$3 = \ln(x) + 1$$

$$2 = \ln(x)$$

$$x = e^2$$

Point:

$$y(e^2) = e^2 \ln(e^2)$$

$$y(e^2) = 2e^2$$

$$(e^2, 2e^2)$$

Eqⁿ:

$$y = mx + b$$

$$2e^2 = 3(e^2) + b$$

$$2e^2 - 3e^2 = b$$

$$b = -e^2$$

$$y = 3x - e^2$$

7) Find all local extrema for $y = \frac{1}{2}x(2)^{3x+1}$.

$$y' = \frac{1}{2}(2)^{3x+1} + \ln(2)(2)^{3x+1}(3)\left(\frac{1}{2}x\right)$$

$$y' = \frac{1}{2}(2)^{3x+1} [1 + 3x \ln(2)]$$

$$0 = \frac{1}{2}(2)^{3x+1} [1 + 3x \ln(2)]$$

$$0 = 2^{3x+1} \quad 0 = 1 + 3x \ln(2)$$

No solution

$$x = \frac{-1}{3 \ln(2)}$$

$$x \approx -0.48$$

$$y(-0.48) \approx -0.18$$

1st deriv. test:

	$-\infty$	-0.48	∞
y'	$-$	$+$	
y	$\searrow$	$\nearrow$	

Min at $(-0.48, -0.18)$

8) Continuous growth or decay follows the formula $A = ce^{kt}$, where c is the initial amount, and k is a rate factor. The mass of a radioactive substance is 1000 g on day 1, and only 100 g after 100 days. Find ...

a) k , then write the equation with c and k

$$100 = 1000 e^{k(100)}$$

$$\frac{1}{10} = e^{100k}$$

$$\ln\left(\frac{1}{10}\right) = 100k$$

$$k = \frac{\ln\left(\frac{1}{10}\right)}{100}$$

$$k \approx -0.023$$

$$A(t) = 1000 e^{-0.023t}$$

b) the half-life,

$$500 = 1000 e^{-0.023t}$$

$$\frac{1}{2} = e^{-0.023t}$$

$$\ln\left(\frac{1}{2}\right) = -0.023t$$

$$t \approx 30.14 \text{ days}$$

b) the amount that remains after 300 days, and

$$A(300) = 1000 e^{-0.023(300)}$$

$$A(300) \approx 1 \text{ gram.}$$

c) the rate of decay after 50 days.

$$A'(t) = 1000 e^{-0.023t} (-0.023)$$

$$A'(t) = -23 e^{-0.023t}$$

$$A'(50) = -23 e^{-0.023(50)}$$

$$A'(50) \approx -7.28 \text{ g/day.}$$

Answers:

1)a) $y = e^{x \ln b}$ b) $\frac{dy}{dx} = (e^{x \ln b}) \ln b$

2)a) $y' = -3e^{-3x}$ b) $f'(x) = 4e^{4x-5}$ c) $y' = 2(e^{2x} + e^{-2x})$ d) $y' = 2^x(\ln 2) + 3^x(\ln 3)$

e) $f'(x) = 6e^{2x} - 3(2^{3x}) \ln 2$ f) $y' = 4xe^x + 4e^x$ g) $y' = -(5^x)(e^{-x})(1 - \ln 5)$ h) $f'(x) = e^{2x}(2x + 1 - 6e^{-5x})$

3)a) $y' = e^{-x}(\cos x - \sin x)$ b) $y' = -\sin x (e^{\cos x})$ c) $f'(x) = e^{2x}(2x^2 - 4x + 1)$ d) $g'(x) = -4xe^{\cos(2x)}[x \sin(2x) - 1]$

4) local max of $y = 0.25$ when $x = \ln(0.5)$

5) $y = 6x + 3$

6) $y = 3x - e^2$

7) $CN \sim -0.48$, so the point $(-0.48, -0.18)$ is a local minimum

8)a) $k \sim -0.023$, so the formula is $A = 1000e^{-0.023t}$

b) $t \sim 30$ days

c) $A(300) \approx 1g$

d) $A'(50) \approx -7.3g/day$